



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SREE PRASANNA VENKATACHALAPATY FUEL

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.

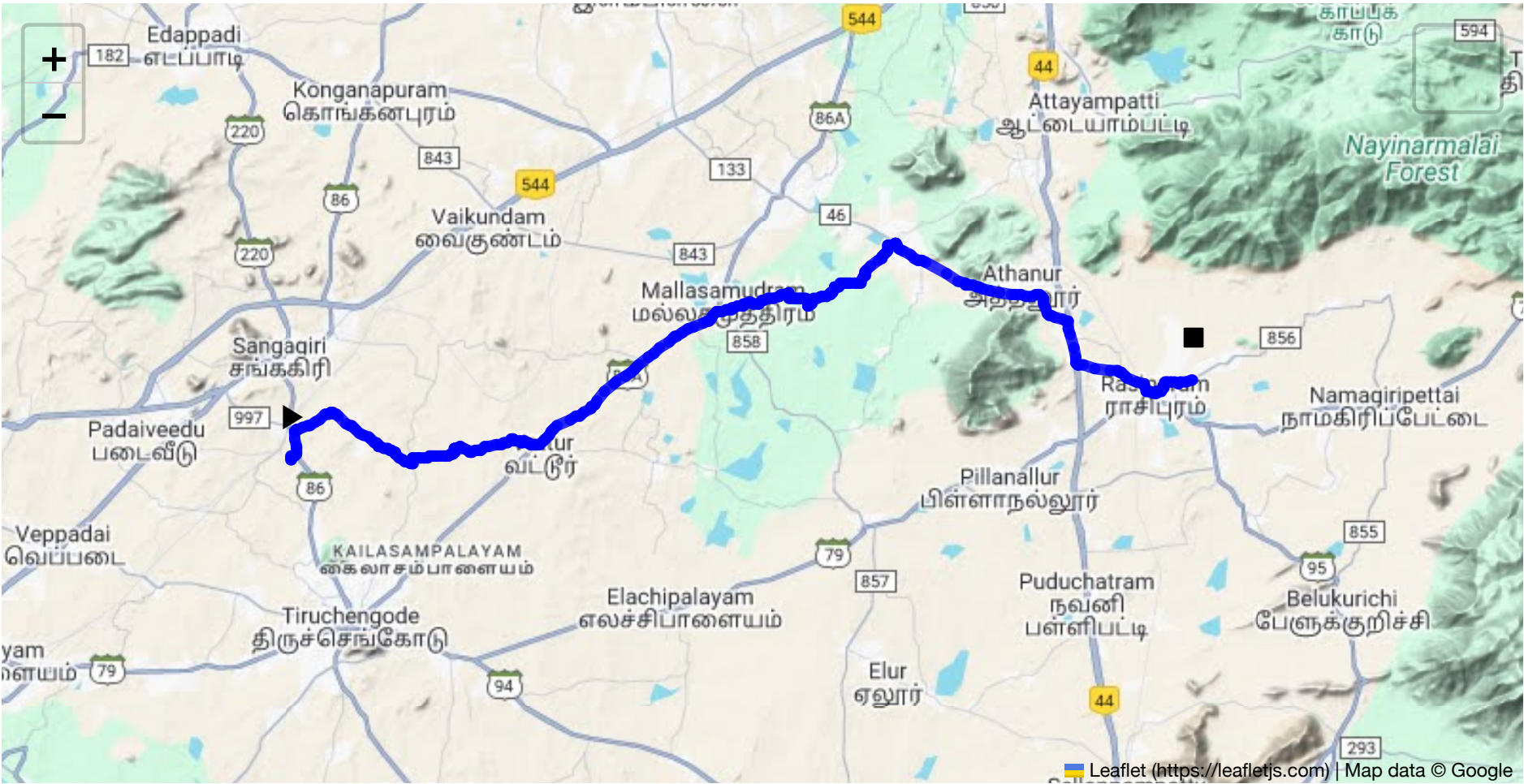


Route Summary:
Total Distance: 45.52 km
Estimated Duration: 1.3 hours

Adjusted Duration (Heavy Vehicle): 1.6 hours

Start: (11.4381, 77.8734)

End: (11.466359, 78.198539)



Welcome to the Journey Risk Management Study

Route Safety Analysis Report

1. Overview of the Route Map: The route begins at CVQF+23W in Sangagiri and concludes at 1, Pattanam Rd, Rasipuram, passing through CXV7+RR7, Morepalayam and Sembampalayam. This spans approximately 45.52 kilometers, primarily following regional roadways that can be narrow and involve rural driving conditions. The driving time for heavy vehicles carrying hazardous materials is around 1.5 hours under normal conditions.

2. Typical Weather Conditions and Potential Weather-Related Hazards: Tamil Nadu generally experiences tropical weather, with hot summers and significant monsoon rains from June to September. During the monsoon, road conditions can deteriorate, leading to slippery surfaces and potential flooding in low-lying areas. Pre-trip planning during the monsoon should include checking forecasts for heavy rain to avoid hydrological hazards.

3. Analysis of Traffic Patterns: Traffic is typically moderate outside of urban centers, with peak congestion during morning (8:00 AM - 10:00 AM) and evening (6:00 PM - 8:00 PM) times in densely populated areas. Notably, traffic congestion is expected near markets and schools, especially in towns like Rasipuram and Sangagiri.

4. Assessment of Road Quality and Infrastructure: The route comprises mainly regional roads that may lack maintenance. Rural roads may have potholes and narrow bridges, potentially impacting vehicle stability and safety. Pay attention to uneven surfaces and ensure speeds are reduced in these areas.

5. Suggestions for Alternative Routes in Emergencies: For emergencies, a useful alternative is to divert via NH44 (National Highway 44), which runs parallel and offers superior road conditions and infrastructure. This detour, while slightly longer, provides better emergency support and transport facilities.

6. Summary of Local Regulations Affecting Hazardous Material Transport: Transporters must adhere to the Central Motor Vehicles Act and Tamil Nadu-specific regulations, which include permits for hazardous material transport, adherence to specified routes, and timing restrictions to reduce exposure to highly populated areas during peak times.

7. Overview of Historical Incidents: There have been reports of traffic accidents in the area, usually due to inadequate road signage and vehicle breakdowns in narrow lanes. Historical incidents typically involve smaller vehicles rather than trucks, but caution is advised given the potential for rural roads to exacerbate accident severity.

8. Environmental Considerations and Sensitive Areas: The route traverses areas near agricultural lands, requiring caution to prevent spills. There are no designated wildlife sanctuaries along this route, but preserving agricultural integrity is paramount due to economic dependence on farming in these communities.

9. Analysis of Communication Coverage: Cellular coverage is generally stable around Sangagiri and Rasipuram, but some rural stretches may have coverage gaps. It is advisable to carry a satellite phone or a robust emergency communication device for consistent connectivity.

10. Estimated Emergency Response Times: In populated areas like Sangagiri and Rasipuram, emergency response times could average 15-30 minutes, influenced by time of day and traffic. In more rural zones or during nighttime, response times may extend to an hour or more. Ensuring a detailed emergency plan is key, including contact numbers of the nearest hospitals and police stations.

12. Overall Summary of Risk Assessment: This route involves moderate risk for transporting hazardous materials, mainly due to variable road conditions, potential weather impacts, limited emergency response capabilities, and the need for compliance with local regulations. It is essential to maintain a cautious driving approach, prepare for weather variations, and ensure all permits and emergency protocols are in place.

By following these recommendations, the journey can be executed safely and efficiently, minimizing risk to both the vehicle and surrounding communities.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.46154, 78.18539	15 KM/Hr
1	Turn	Medium	11.43822, 77.87348	30 KM/Hr
2	Turn	High	11.43968, 77.87345	15 KM/Hr
3	Turn	High	11.44029, 77.87544	15 KM/Hr
4	Blind Spot	Blind Spot	11.44861, 77.87429	10 KM/Hr
5	Turn	Medium	11.45478, 77.88925	30 KM/Hr
6	Turn	High	11.43675, 77.91693	15 KM/Hr
7	Turn	High	11.43684, 77.91706	15 KM/Hr
8	Turn	High	11.43926, 77.91705	15 KM/Hr
9	Turn	Medium	11.43930, 77.92282	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
10	Turn	Medium	11.43967, 77.92345	30 KM/Hr
11	Turn	Medium	11.43979, 77.93009	30 KM/Hr
12	Turn	Medium	11.44030, 77.93066	30 KM/Hr
13	Turn	Medium	11.44067, 77.93203	30 KM/Hr
14	Turn	High	11.44088, 77.93223	15 KM/Hr
15	Turn	Medium	11.44127, 77.93218	30 KM/Hr
16	Turn	Medium	11.44163, 77.93244	30 KM/Hr
17	Turn	High	11.44243, 77.93244	15 KM/Hr
18	Turn	Medium	11.44285, 77.93440	30 KM/Hr
19	Turn	Medium	11.44093, 77.93821	30 KM/Hr
20	Turn	Medium	11.44550, 77.95329	30 KM/Hr
21	Turn	Medium	11.44369, 77.96036	30 KM/Hr
22	Turn	Medium	11.44395, 77.96298	30 KM/Hr
23	Turn	High	11.49063, 78.02680	15 KM/Hr
24	Turn	High	11.49135, 78.03252	15 KM/Hr
25	Turn	High	11.49183, 78.03257	15 KM/Hr
26	Turn	Medium	11.49355, 78.03703	30 KM/Hr
27	Turn	Medium	11.49410, 78.03864	30 KM/Hr
28	Turn	Medium	11.49381, 78.04241	30 KM/Hr
29	Turn	Medium	11.49637, 78.04986	30 KM/Hr
30	Turn	High	11.49676, 78.05010	15 KM/Hr
31	Blind Spot	Blind Spot	11.49723, 78.05308	10 KM/Hr
32	Turn	High	11.49553, 78.05338	15 KM/Hr
33	Turn	Medium	11.49547, 78.05442	30 KM/Hr
34	Turn	High	11.49518, 78.05498	15 KM/Hr
35	Turn	Medium	11.49526, 78.05677	30 KM/Hr
36	Blind Spot	Blind Spot	11.49548, 78.05904	10 KM/Hr
37	Turn	High	11.49469, 78.05908	15 KM/Hr
38	Turn	High	11.49451, 78.05995	15 KM/Hr
39	Turn	High	11.49314, 78.05979	15 KM/Hr
40	Turn	High	11.49305, 78.05987	15 KM/Hr
41	Turn	Medium	11.49306, 78.06011	30 KM/Hr
42	Turn	Medium	11.49358, 78.06081	30 KM/Hr
43	Turn	High	11.49370, 78.06142	15 KM/Hr
44	Turn	High	11.49561, 78.06192	15 KM/Hr
45	Turn	High	11.49665, 78.06789	15 KM/Hr
46	Turn	High	11.49725, 78.06787	15 KM/Hr
47	Turn	Medium	11.49738, 78.06806	30 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
48	Turn	High	11.49738, 78.06877	15 KM/Hr
49	Turn	High	11.49906, 78.06934	15 KM/Hr
50	Turn	Medium	11.49969, 78.07116	30 KM/Hr
51	Turn	High	11.50118, 78.07188	15 KM/Hr
52	Turn	Medium	11.50070, 78.07343	30 KM/Hr
53	Turn	Medium	11.50092, 78.07374	30 KM/Hr
54	Turn	Medium	11.50085, 78.07854	30 KM/Hr
55	Turn	Medium	11.50143, 78.07938	30 KM/Hr
56	Turn	High	11.50374, 78.08048	15 KM/Hr
57	Turn	High	11.50424, 78.08040	15 KM/Hr
58	Turn	Medium	11.51303, 78.08909	30 KM/Hr
59	Blind Spot	Blind Spot	11.51452, 78.08915	10 KM/Hr
60	Turn	High	11.51417, 78.09091	15 KM/Hr
61	Turn	Medium	11.51445, 78.09107	30 KM/Hr
62	Turn	Medium	11.51468, 78.09148	30 KM/Hr
63	Blind Spot	Blind Spot	11.51494, 78.09165	10 KM/Hr
64	Turn	Medium	11.51485, 78.09173	30 KM/Hr
65	Turn	Medium	11.48976, 78.14718	30 KM/Hr
66	Turn	High	11.48720, 78.15384	15 KM/Hr
67	Turn	High	11.47225, 78.15722	15 KM/Hr
68	Turn	High	11.46504, 78.18145	15 KM/Hr
69	Turn	Medium	11.46285, 78.18233	30 KM/Hr
70	Turn	Medium	11.46200, 78.18554	30 KM/Hr
71	Turn	Medium	11.46228, 78.18584	30 KM/Hr
72	Turn	High	11.46266, 78.18733	15 KM/Hr
73	Turn	High	11.46329, 78.18764	15 KM/Hr
74	Turn	Medium	11.46342, 78.18797	30 KM/Hr
75	Turn	High	11.46594, 78.18897	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
3	hospital	Government Hospital, Vennandur	11.51701, 78.088631	30 km/h	Medium
5	hospital	surya hospital	11.4622163, 78.1840544	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
6	police	Police Station Rasipuram	11.4617825, 78.1847411	30 km/h	Medium
7	hospital	Government Hospital Blood Bank, Rasipuram	11.4612082, 78.1849249	30 km/h	Medium
8	hospital	dr.thangavel dch hospital	11.4619955, 78.1849181	30 km/h	Medium
9	hospital	Government Hospital, Rasipuram	11.4612987, 78.1851369	30 km/h	Medium
10	hospital	Rasi Hospital	11.4618368, 78.1855695	30 km/h	Medium
11	hospital	M.R. Hospital, Namakkal	11.4613462, 78.1861702	30 km/h	Medium
14	hospital	Aravind Hospital	11.46478, 78.1925941	30 km/h	Medium
15	hospital	Sri Venkatesh Eye Care	11.4649052, 78.1928421	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	college	Mahendra Engineering College	11.4770412, 78.0024214	30 km/h	Medium
1	school	Akv Schools	11.4850407, 78.0227236	30 km/h	Medium
2	school	Akv Schools	11.4854328, 78.0233558	30 km/h	Medium
4	school	SRV School	11.4688166, 78.1634582	30 km/h	Medium
12	school	Government Hr Sec School	11.4654519, 78.1869866	30 km/h	Medium
13	school	Government Hr Sec School	11.4654357, 78.186985	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46154, 78.18539



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.44861, 77.87429



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45478, 77.88925



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43675, 77.91693



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43684, 77.91706



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.43926, 77.91705



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43930, 77.92282



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43967, 77.92345



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.43979, 77.93009



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44030, 77.93066



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44067, 77.93203



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44088, 77.93223



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44127, 77.93218



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44163, 77.93244



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44243, 77.93244



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44285, 77.93440



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44093, 77.93821



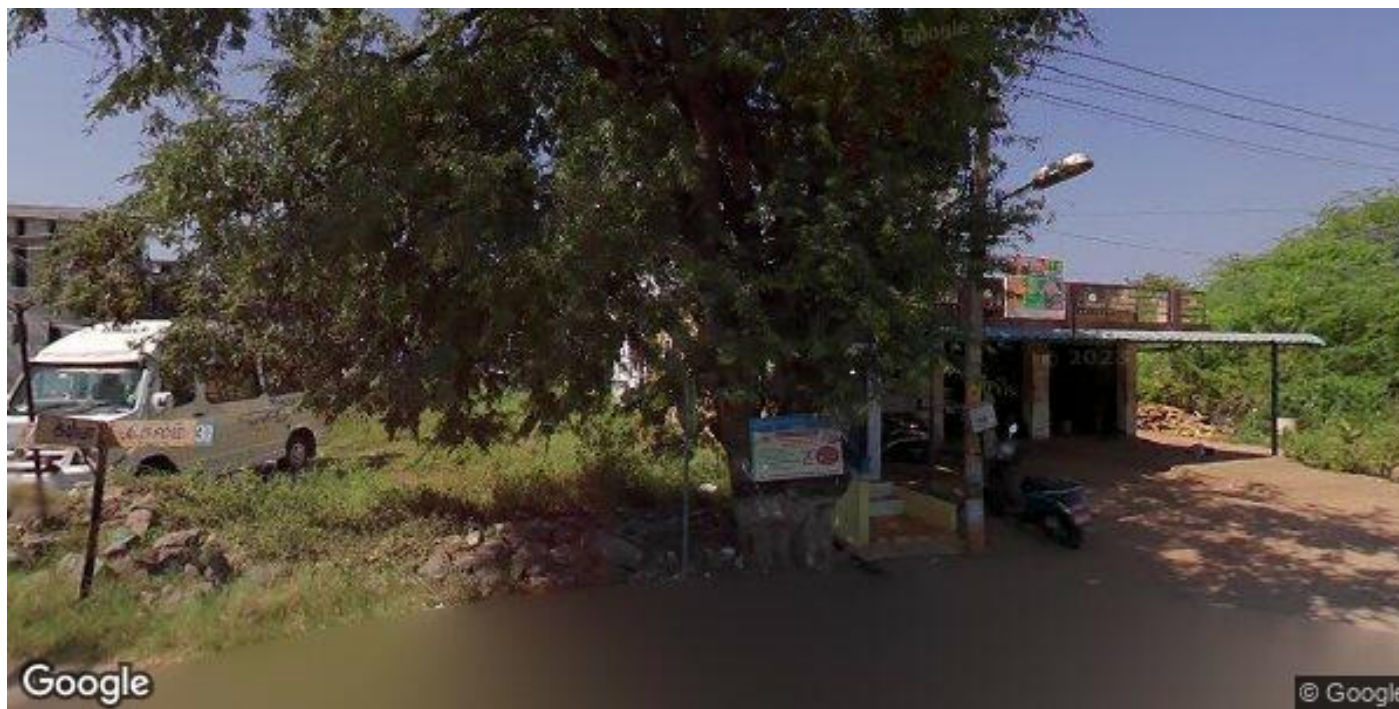
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44550, 77.95329



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44369, 77.96036



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44395, 77.96298



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49063, 78.02680



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49135, 78.03252



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49183, 78.03257



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49355, 78.03703



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49410, 78.03864



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49381, 78.04241



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49637, 78.04986



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49676, 78.05010



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49723, 78.05308



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49553, 78.05338



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49547, 78.05442



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49518, 78.05498



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49526, 78.05677



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49548, 78.05904



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49469, 78.05908



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49451, 78.05995



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49314, 78.05979



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49305, 78.05987



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49306, 78.06011



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49358, 78.06081



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49370, 78.06142



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49561, 78.06192



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49665, 78.06789



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49725, 78.06787



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49738, 78.06806



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49738, 78.06877



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.49906, 78.06934



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49969, 78.07116



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50118, 78.07188



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50070, 78.07343



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50092, 78.07374



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50085, 78.07854



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50143, 78.07938



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50374, 78.08048



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.50424, 78.08040



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51303, 78.08909



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51452, 78.08915



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.51417, 78.09091



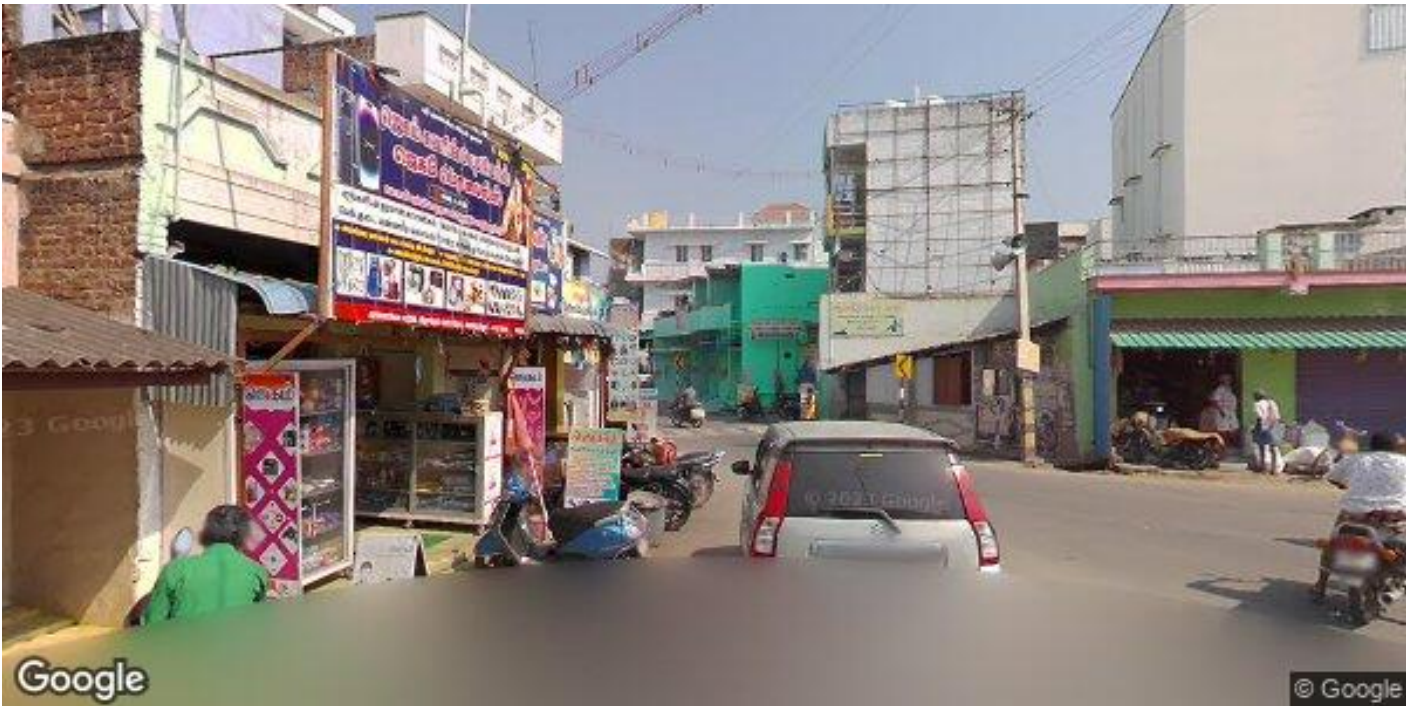
Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51445, 78.09107



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51468, 78.09148



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.51494, 78.09165



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.51485, 78.09173



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.48976, 78.14718



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.48720, 78.15384



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.47225, 78.15722



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46504, 78.18145



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46285, 78.18233



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46200, 78.18554



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46228, 78.18584



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46266, 78.18733



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46329, 78.18764



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46342, 78.18797



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.46594, 78.18897

